TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Specific Heat of a Solid by the Method of Mixture

Heat a metal block in steam, drop it into cool water, and let the water's small rise in temperature report how much heat each gram of the metal was carrying.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 12 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the specific heat s of a solid (here a brass cylinder) by the method of mixtures. The solid is heated to a known high temperature θ_2 in a steam heater (Regnault's apparatus), then transferred quickly into a calorimeter holding water at θ_1. It cools to the common final temperature θ, surrendering heat

$$\text{heat lost by solid} = m_s\, s\, (\theta_2 - \theta)$$

which is received by the water and by the calorimeter vessel, stirrer and thermometer, whose combined appetite is again their **water equivalent** W:

$$\text{heat gained} = (m_w + W)(\theta - \theta_1) \quad \Rightarrow \quad s = (m_w + W)(\theta - \theta_1) \,/\, (\, m_s\, (\theta_2 - \theta)\,)$$

The whole method is bookkeeping: every calorie the solid loses must appear in the water or the vessel. The water equivalent is found from the calorimeter's own mass and material — a copper vessel of mass M has $W = M \times s_{Cu}$, with $s_{Cu} \approx 0.09$ cal g^{-1} $°C^{-1}$ from the tables. Steam heating is chosen because condensing steam holds the solid at one known, steady temperature — the steam point — with no flame to overshoot it.

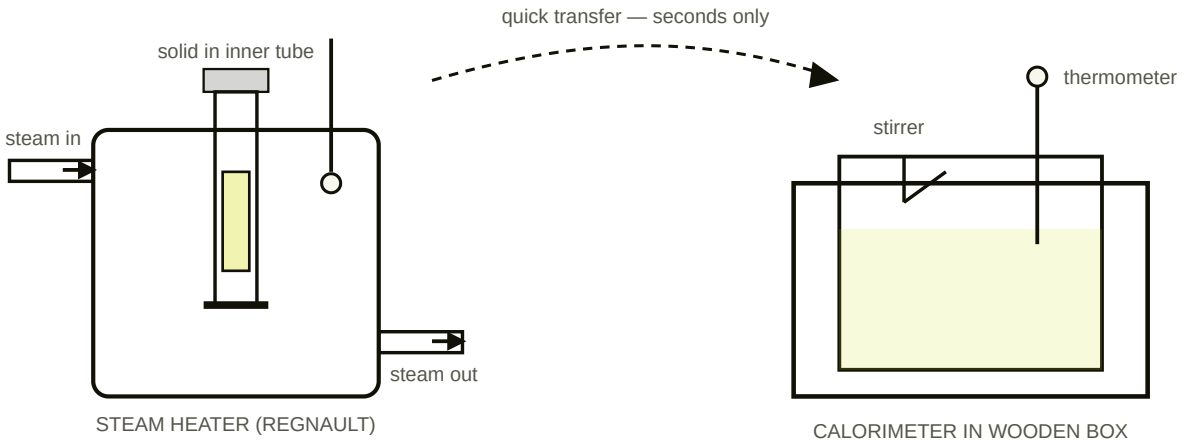


Figure 1 — Regnault's steam heater and the calorimeter. The solid soaks to the steam temperature inside the heater's inner tube; the shutter at the tube's foot then opens and the solid drops straight into the calorimeter, spending only a second or two in the air.

2. Apparatus and procedure

- **Apparatus:** Regnault's steam heater, calorimeter with stirrer in a wooden box, thermometer (0.1 °C), balance, steam generator, the solid (brass cylinder with a thread).
- Weigh the solid, m_s. Weigh the empty calorimeter with stirrer (needed for W); add a measured mass of water m_w — enough to cover the solid completely once it is dropped in.
- Suspend the solid in the heater's inner tube and pass steam until the thermometer in the heater is steady at the steam temperature θ_2 (about 10 minutes — the solid must be hot all through, not just on its skin).
- Meanwhile stir the calorimeter water and record its temperature θ_1 just before the transfer.
- Move the calorimeter under the heater, open the shutter, and let the solid fall in — one quick, practised motion. Replace the lid, stir gently, and record the highest temperature reached, θ.
- Compute s from the mixture equation. (For the finest work a small cooling correction is added to θ, since the mixture stands a little above room temperature and leaks heat while settling.)

3. Worked calculation

Brass solid: $m_s = 100$ g, heated in steam at $\theta_2 = 100$ °C. Water: $m_w = 150$ g at $\theta_1 = 30$ °C. Calorimeter: copper, mass 110 g, so $W = 110 \times 0.09 \approx 10$ g. Final temperature $\theta = 33.7$ °C.

$$\text{heat gained} = (150 + 10)(33.7 - 30) = 160 \times 3.7 = 592 \text{ cal}$$

$$s = 592 / (100 \times (100 - 33.7)) = 592 / 6630 = 0.089 \text{ cal g}^{-1}\,^{\circ}\text{C}^{-1}$$

In SI units, $s \approx 0.089 \times 4180 \approx 3.7 \times 10^2$ J kg^{-1} K^{-1}, against the accepted value for brass of about 0.09 cal g^{-1} °C^{-1} (380 J kg^{-1} K^{-1}). Notice how small the water's rise is — 3.7 °C against the solid's fall of 66.3 °C: a metal carries little heat per gram compared with water.

4. Precautions

- Speed and dryness at the transfer: the solid cools in air and can carry clinging hot water from the heater. A slow transfer understates θ_2; carried-over water adds heat that the solid never supplied — both falsify the bookkeeping.
- The water must cover the solid fully, or part of its heat escapes to the air instead of the water.
- Stir gently and read the *maximum* temperature — after the peak, radiation losses pull the reading down and θ comes out low.
- Keep the temperature rise small (a few degrees): the bigger the mixture's excess over room temperature, the faster it leaks heat while you watch it.

Practice (viva preparation)

1. Why is the solid heated in steam rather than directly in a flame or in boiling water on a burner?
2. A 50 g metal block at 100 °C is dropped into 100 g of water at 25 °C in a calorimeter of water equivalent 8 g. The final temperature is 28 °C. Find the specific heat of the metal.
3. Why is the calorimeter's water equivalent not simply ignored — how large is its share in the worked example?
4. The solid is left out in the air for half a minute during a fumbled transfer. What happens to the computed s, and why?
5. Why should the water in the calorimeter start a little below room temperature if possible?

Answer key

1. Steam condensing on the solid holds it at one uniform, known temperature — the steam point (100 °C) — until it is hot throughout. A flame heats unevenly and to an unknown temperature; the method needs θ_2 to be a fact, not a guess.
2. Heat gained = $(100 + 8)(28 - 25) = 324$ cal. Heat lost = $50 \times s \times (100 - 28) = 3600\,s$. So $s = 324/3600 = 0.090$ cal g^{-1} °C^{-1}.
3. The vessel drinks alongside the water: here it takes $W(\theta - \theta_1) = 10 \times 3.7 = 37$ cal of the 592 cal gained — about 6%. Ignoring it would understate the heat received, and so understate s, by that share.

4. The solid arrives cooler than the recorded θ_2 = 100 °C, so it delivers less heat; the formula still charges it with a fall from 100 °C, and the computed s comes out too small.
5. Below room temperature the calorimeter gains heat from the room; above it, it loses heat. Starting slightly below and ending slightly above makes the two nearly cancel — the same symmetry trick as in Experiment 7.

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